***Solution:***

(i) First convert the given equations $Y$ on $X$ and $X$ on $Y$ in standard form and find their regression coefficients respectively.

Given regression lines are

$3X-2Y = 5$ ... (1)

$X-4Y = 7$ ... (2)

Let the line of regression of $X$ on $Y$ is

$3X-2Y = 5$

$3X = 2Y+5$

$X = \frac{1}{3}(2Y+5)$

$X = \frac{1}{3}(2Y+5)$

$X = \frac{2}{3}Y+\frac{5}{3}$

$\therefore$ Regression coefficient of $X$ on $Y$ is

$b_{xy} = \frac{2}{3}(<1)$

Let the line of regression of Y on X is

$X-4Y = 7$

$-4Y = -X+7$

$4Y = X-7$

$Y = \frac{1}{4}(X-7)$

$Y = \frac{1}{4}X-\frac{7}{4}$

$\therefore$ Regression coefficient of Y on X is

$b_{yx} = \frac{1}{4}(<1)$

(ii) **Coefficient of correlation**

Since the two regression coefficients are positive then the correlation coefficient is also positive and it is given by

$r = \sqrt{b_{yx} \cdot b_{xy}}$

$= \sqrt{\frac{2}{3} \cdot \frac{1}{4}}$

$= \sqrt{\frac{1}{6}}$

$= 0.4082$

$\therefore r = 0.4082$

## Exercise 9.2

1. From the data given below:

| | | | | | |
|---|---|---|---|---|---|
| Marks in Economics: | 25 | 28 | 35 | 32 | 31 |
| Marks in Statistics: | 43 | 46 | 49 | 41 | 36 |
| Marks in Economics: | 36 | 29 | 38 | 34 | 32 |
| Marks in Statistics: | 32 | 31 | 30 | 33 | 39 |

Find (a) The two regression equations, (b) The coefficient of correlation between marks in Economics and statistics, (c) The mostly likely marks in Statistics when the marks in Economics is 30.

2. The heights (in cm.) of a group of fathers and sons are given below:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Heights of fathers: | 158 | 166 | 163 | 165 | 167 | 170 | 167 | 172 | 177 | 181 |
| Heights of Sons : | 163 | 158 | 167 | 170 | 160 | 180 | 170 | 175 | 172 | 175 |

Find the lines of regression and estimate the height of son when the height of the father is 164 cm.

3. The following data give the height in inches ($X$) and the weight in lb. ($Y$) of a random sample of 10 students from a large group of students of age 17 years:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| X | 61 | 68 | 68 | 64 | 65 | 70 | 63 | 62 | 64 | 67 |
| Y | 112 | 123 | 130 | 115 | 110 | 125 | 100 | 113 | 116 | 125 |

Estimate weight of the student of a height 69 inches.

4. Obtain the two regression lines from the following data $N=20$, $\Sigma X=80$, $\Sigma Y=40$, $\Sigma X^2=1680$, $\Sigma Y^2=320$ and $\Sigma XY=480$.



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